

Cubit brief:

- Need compatible SolidWorks model and version (2025 from PSU)
- Need CoreForm Cubit installed via access code request

In SolidWorks:

- Import your parts and put together an assembly, making mates and relations as you see fit for reference geometry
- When you have the assembly as you want it, **export to ACIS** (.SAT format) from solidworks to your machine

In Cubit:

SCALING AND TRANSLATION:

- Scaling:
 - o When initially importing a SolidWorks model, you need to at the very least scale the model down to the centimeter scale.
 - o Cubit will inherit whatever the dominant scale used by SolidWorks is (preferably choose the meter scale for a quick conversion) (multiply by 0.1 scale factor if you exported in “mm-g-s” format on the assembly, since Cubit will look at the base unit of distance = “1”, assign that to single unit mm, and your entire model will be off by that amount
 - o Manual check:
 - Create a volume [Brick] of 100 “units” by “100 units” by “100” units which, if your model is scaled correctly, should appear as a cubic meter box, compare this cubic meter to the dimensions of your model to verify your dimensions are correct
 - For D3D – creating this brick will do so at the center of the Cubit environment
- Translation:
 - o Select your “center” volume and highlight its coordinates when you import it from SolidWorks
 - o Using the coordinates from your model, select all volumes to move BY DISTANCE, the negative of the coordinates of your center piece
 - Moving that amount of distance, applied on all of the components, will move everything offset to the origin of the cubit instance.
 - o Note: OpenMC will assume your model is at center when called, if you don’t perform this step, your actual geometry might be somewhere completely different.

IMPRINT AND MERGE:

- With your geometry imported, run the following commands directly in the command line:
 - o imprint body all
 - o merge body all
- This will have Cubit look for overlapping surfaces between volumes, imprint them on one another, and then merge the two together so that there is a smooth transition between them as seen by OpenMC particles
- Note: Failure to do this may lead to lost particles or longer run times
- Note: If you keep a lot of your geometries separate in SolidWorks, you won't need to worry about this as much

MATERIALS:

- Making Materials (assigns a name to things)
 - o Go to the material tab, and create new materials with appropriate names, this is just to give you a menu to select later
 - o Note: The names provided to Cubit should match the material names used in OpenMC, they may or may not be case sensitive, but treat it like its case sensitive.
 - o This is just to give OpenMC something to look at on the material file when it tracks particles in a volume.
- Graveyard material
 - o If you are not dealing with a reflective boundary condition problem (like an MEPD fuel element), you don't need to create a graveyard material, However, for D3D you do need a graveyard material called "graveyard" (exact).
- Implicit complement (Optional)
 - o There are commands to aid a specific material (once created) to graveyard from command line to fill with something (it's the space not occupied by a volume)
 - I don't have these on hand, you'll need to find them but there is a command to add air to the implicit complement

VOLUME HANDLING:

- Around your geometry, (for the D3D model at least):
 - o Need a graveyard finite-thickness volume set up
 - o Create this by making two bricks around the volume, subtracting the smaller of the two from the larger, and then perform material / block assignment as normal.

BLOCK ASSIGNMENTS:

- Select multiple volumes (that you want to have the same material assigned to), and assign these volumes to single blocks
 - o You can check to see if you've assigned them correctly by selecting the block on the left side of the UI and making sure the volumes highlighted are assigned to that block

BLOCK – MATERIAL FILL:

- With your blocks set up, you will then need to fill each individual block by assigning materials to the blocks, which can be done using the right-hand menu tab
 - o There isn't a good way to check that you've done this correctly, you kinda just have to be careful with it.

SURFACE MESHING:

- On block assignment, you'll need to then mesh the

EXPORT:

- With mesh created and materials assigned export to DAGMC (.h5m) format (can do this from command line, though I prefer to use the GUI)

Cubit notes:

- Cubit doesn't seem to care what the order of things you do is, but its probably best to imprint and merge as soon as you can.

OpenMC:

- Setup DAGMC universe
- Materials defined in the material xml file need to match to anything that the DAGMC file will call
- *Hypothetically* to get around Cubit's entity mesh limit (50,000), you might be able to import multiple dagmc instances over each other (with equal implicit complements) which OpenMC would hopefully read as just present and overlapping

Troubleshooting:

Cubit:

- Export fail (exceeds 50,000 elements):
 - o Lower the number of meshes used in the model to below 50,000 or attempt to import other meshes over one another in OpenMC

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OpenMC/DAGMC:

- Signal 11 Segmentation fault:
 - Root cause: occurs due to a memory accessing error
 - DagMC cause: DAGMC universe calls a material file that can't be read in the XML
 - Bootstrap fix -> rename the XML material to what DAGMC sees in its material list (D3D example: Concrete-Bentonite mix is a mixture, but has a material name different from constituent parts, you can rename the material XML something different to handle this)
 - Long-term fix -> in Cubit, rename the material to what OpenMC will write the material as by default (important for mixing isotopes)
 - DagMC cause: DAGMC universe has overlapping geometries
 - Fix -> Review the geometry in Cubit with built in overlap functions / intersection checks, if possible, separate volumes in cubit manually by offsetting them
 - Fix -> Go to solidworks assembly and visually inspect for overlaps, re-modify geometry to work appropriately.
 - **This may not be a segmentation fault, could appear as a cell error**